

A Statistical Perspective of Non- Inferiority

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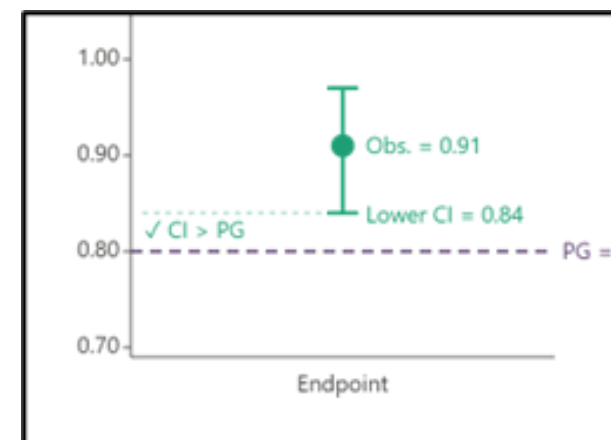
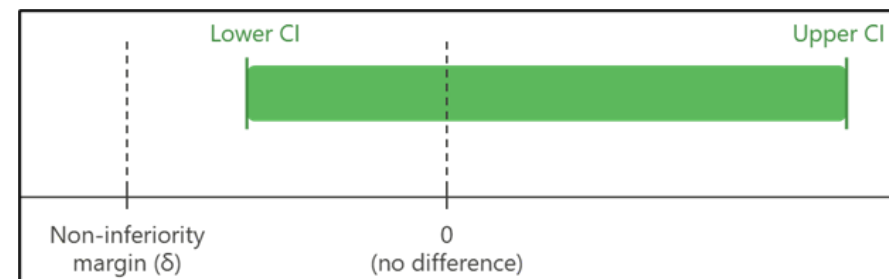
Section 1

- Discuss Non-Inferiority & Performance Goal Studies
- Describe our Case Study
- Determine Performance Endpoints

Non-Inferiority RCT vs Performance Goal Studies

Aim: To show that the new treatment is no worse in safety & efficacy than current standard of care.

1. **Randomized Control Trial:** A 2-arm trial between a new treatment & active control where the new treatment & confidence interval must be within the **non-inferiority margin** of the control's performance.
2. **Performance Goal Study:** A single-arm trial where results are benchmarked against predefined performance goals. Typical for medical devices as blinding is impractical and/or an active control is financially unjustifiable.



How are Performance Goals Calculated?

Performance Goals define the **worst acceptable outcome** a new device must beat to be approved.

Process:

1. Identify comparable approved devices
2. Get their safety & efficacy performances from their clinical studies
3. Propose these PGs to regulatory authorities (FDA) for approval
4. Authorities can accept, reject, or modify based on clinical rationale

Safety: Maximum acceptable harm

Efficacy: Minimum acceptable clinical benefit preserved

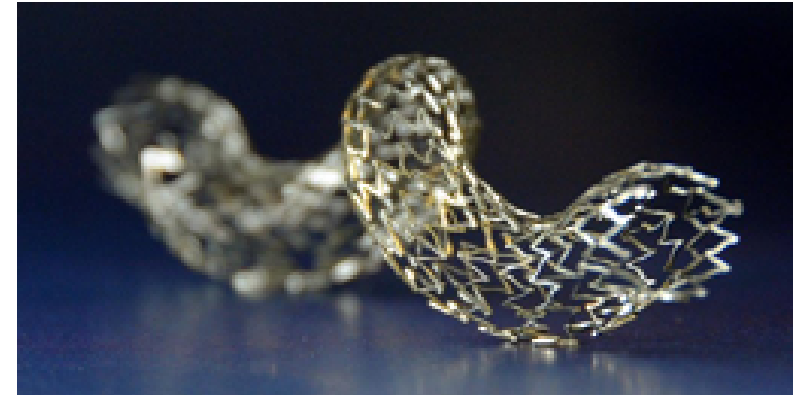
Case Study:

The BioMimics 3D Stent Study:

A pivotal single-arm performance goal study for a peripheral vascular stent (to treat peripheral vascular disease).

Goal:

1. Propose performance goal endpoints for safety & efficacy to show to regulatory authorities.
2. Demonstrate that the new treatment stays below/above the approved performance goals.



😊 If both performance goals are met – the treatment successfully demonstrates non-inferiority 😊

Proposed Performance Goal

Endpoints:

Step 1. Conduct a targeted literature review of clinical trials of comparable stents.

Step 2. Conduct an aggregate-level meta-analysis using a 2007 guidance document of FDA-approved stents.

‘Performance Goals and Endpoint Assessments for Clinical Trials of Femoropopliteal Bare Nitinol Stents in Patients with Symptomatic Peripheral Arterial Disease’

Includes three studies of similar stents assessing the primary endpoints for safety & efficacy.

Proposed Endpoints:

Safety (30-day): Maximum acceptable level of harm

- Amputation
- Death
- Target Vessel Revascularisation (TVR)
- **Primary composite (Combined amputation, death, TVR)**

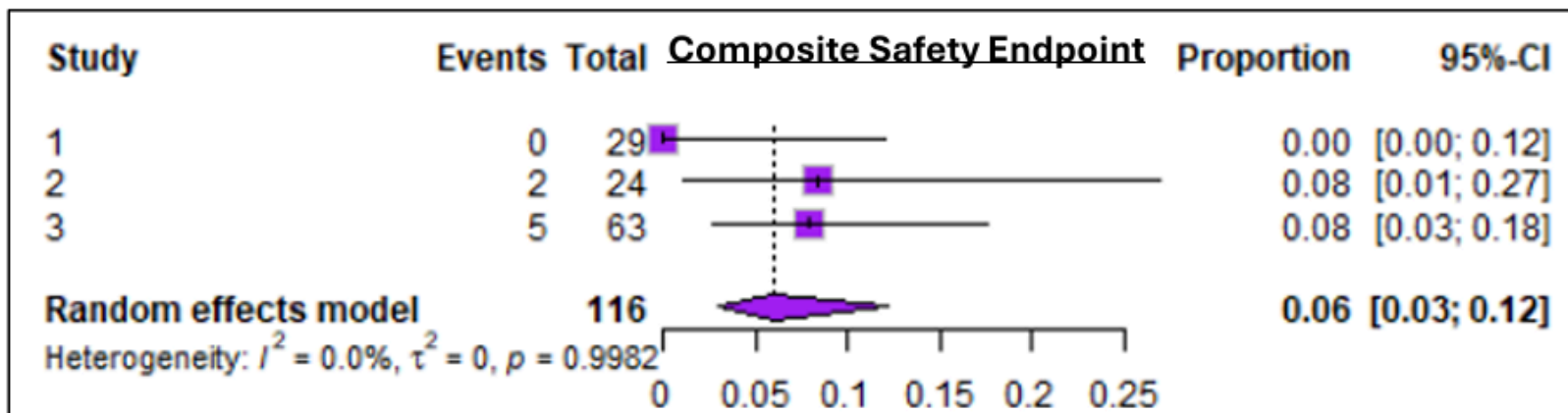
Efficacy (12-month): Minimum clinical benefit that must be preserved

- **Rutherford Classification Change (improved/stable, worsened by x1, worsened by x2)**

Meta Analysis:

A logit-transformation was applied (*extreme proportions*), variance was pooled using a generalised linear mixed model (*robust with small sample sizes*), with study-level variances drawn from each study's observed proportion (*not assuming homogeneity across studies*).

- The upper/lower confidence interval bounds — not point estimate — were proposed to the regulators to account for the uncertainty.
- Note: With only 3 studies available, confidence intervals are wide (conservative)



Performance Goals & Decision Criteria

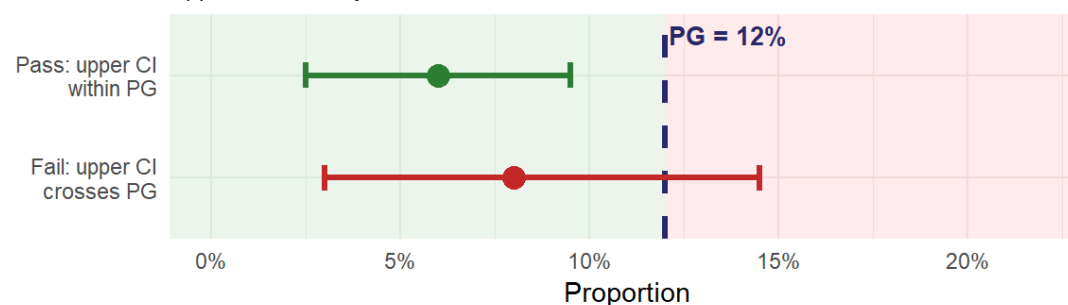
Table 1

Endpoint	Pooled Rate (95% CI)	Performance Goal
Safety — 30 Days		
Composite (Death, Amputation, TVR)	6.0% [2.9–12.1%]	$\leq 12\%$
Efficacy — 12 Months		
RCC Score 1: Improved / No Change	95.8% [87.9–98.6%]	$\geq 88\%$

NI Decision: CI must fall in the green zone for both endpoints

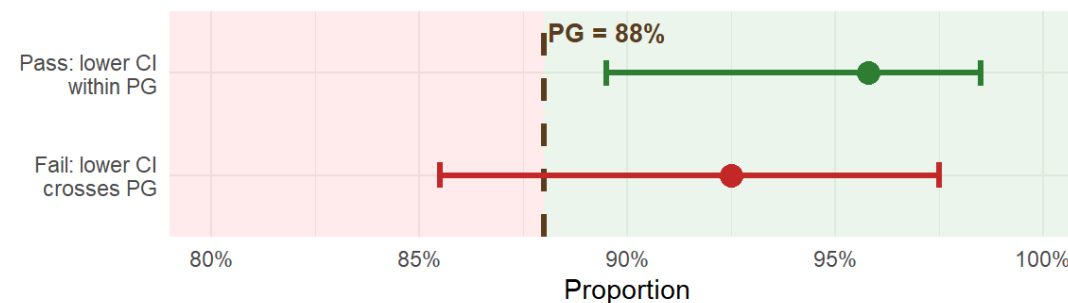
Safety — 30-Day Composite

Upper CI must stay below 12%



Efficacy — RCC Improved/No Change (12 months)

Lower CI must stay above 88%



Section 2

- Explore CIs for Proportions in PG scenarios
- Assess CI performance using a Simulation Study
- Examine CI's impact on Sample Size
- Present **PG-Power**: An interactive Shiny App
- Discuss Final Steps

Why Confidence Interval Method Matters

- The chosen CI method affects the required sample size to achieve adequate power.
- It must be pre-specified in the Statistical Analysis Plan (SAP)
- **Companies:** Larger sample sizes increase study costs, resources, and timelines.
- **Patients:** Larger samples sizes may delay device regulatory approval and patient access to beneficial technology



The Challenge for Confidence Intervals

- CI struggle when proportions are extreme.

Low Complication: Observed proportions close to 0

High Success: Observed proportions close to 1



Wald's Confidence Interval

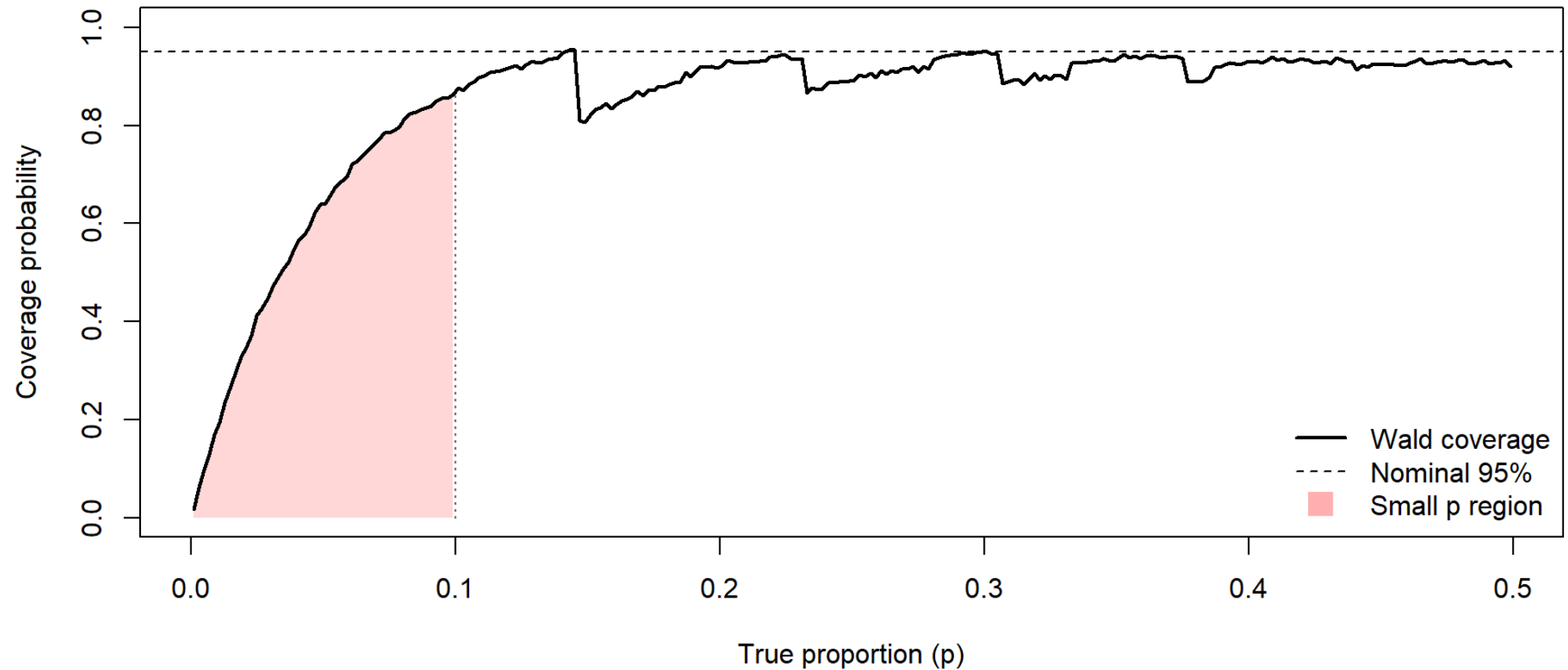
$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- **Limitations:**

- Assumes $\hat{p}$ follows a normal distribution
- Can produce impossible CI bounds outside the $[0,1]$
- Underestimates uncertainty at boundaries
- Results in poor coverage, below nominal 95%



Wald Interval: Coverage Probability Across True Proportions



Wald interval coverage across a range of true proportions when $n = 20$. For small values of p (shaded region), coverage falls well below 95%.

Alternatives

- Wilson Score Interval
- Wilson Score Interval (*with continuity correction*)
- Agresti–Coull Interval
- Clopper–Pearson Interval
- Jeffreys Interval

Wilson Score Interval

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

- Inverts score test for a binomial proportion
- Identifies values of the true proportion p consistent with the observed data
- Doesn't rely on normal approximation → Improves Coverage

- Wald deems estimate to be center of interval
- Wilson fixes that with:

$$\tilde{p} = \frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}}$$

Adjusting center of interval

$$\frac{z^2}{2n}$$

Pulls centre away from 0 or 1

Wilson Score Interval with Continuity Correction

Derived by inverting the score test: $\frac{(\hat{p} - p)^2}{\hat{p}(1 - \hat{p})/n}$

- **Prop.test() (R)**
 - Applies a continuity correction
 - Adjusts the score test boundary by (0.5)
 - Produces wider, more conservative intervals

Agresti-Coull Interval

Modifies the Wald approach by adding pseudo-observations stabilising the estimate.

- Adding 2 successes and 2 failures (for a 95% CI)

$$\hat{p} = \frac{x}{n} \quad \longrightarrow \quad \tilde{p} = \frac{x + 2}{n + 4}$$

Then calculates Wald CI:

$$\tilde{p} \pm 1.96 \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}}$$

Clopper-Pearson Interval

- Based on **exact** binomial distribution

$$X \sim \text{Binomial}(n, p)$$

- The interval is obtained by inverting the exact binomial test:

$$\left[\text{Beta}^{-1}\left(\frac{\alpha}{2}; x, n - x + 1\right), \text{Beta}^{-1}\left(1 - \frac{\alpha}{2}; x + 1, n - x\right) \right]$$

Keep values of p that would **not be rejected** by the observed data

Guarantees nominal coverage (often slightly conservative)



Jeffrey Interval

A Bayesian credible interval for a binomial proportion:

$$X \sim \text{Binomial}(n, p), \quad p \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$$

Starts with a weak prior and updates with a beta posterior. The interval obtained becomes:

$$\left[\text{Beta}^{-1}\left(\frac{\alpha}{2}; x + \frac{1}{2}, n - x + \frac{1}{2}\right), \text{Beta}^{-1}\left(1 - \frac{\alpha}{2}; x + \frac{1}{2}, n - x + \frac{1}{2}\right) \right]$$

- Although Bayesian in construction, the Jeffreys interval often shows strong frequentist performance in terms of coverage, particularly in small samples and boundary cases.

Simulation Study

REVSTAT – Statistical Journal
Volume 6, Number 2, June 2008, 165–197

INTERVAL ESTIMATORS FOR A BINOMIAL PROPORTION: COMPARISON OF TWENTY METHODS

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Abstract:

- In applied statistics it is often necessary to obtain an interval estimate for an unknown proportion (p) based on binomial sampling. This topic is considered in almost every introductory course. However, the usual approximation is known to be poor when the true p is close to zero or to one. To identify alternative procedures with better properties twenty non-iterative methods for computing a (central) two-sided interval estimate for p were selected and compared in terms of coverage probability and expected length. From this study a clear classification of those methods has emerged. An important conclusion is that the interval based on asymptotic normality, but after the arcsine transformation and a continuity correction, and the Add 4 method of Agresti and Coull (1998) yield very reliable results, the choice between the two depending on the desired degree of conservativeness.

Key-Words:

- confidence interval; binomial distribution; proportion test; normal approximation; arcsine transformation; continuity correction; bootstrap; HPD credibility intervals.

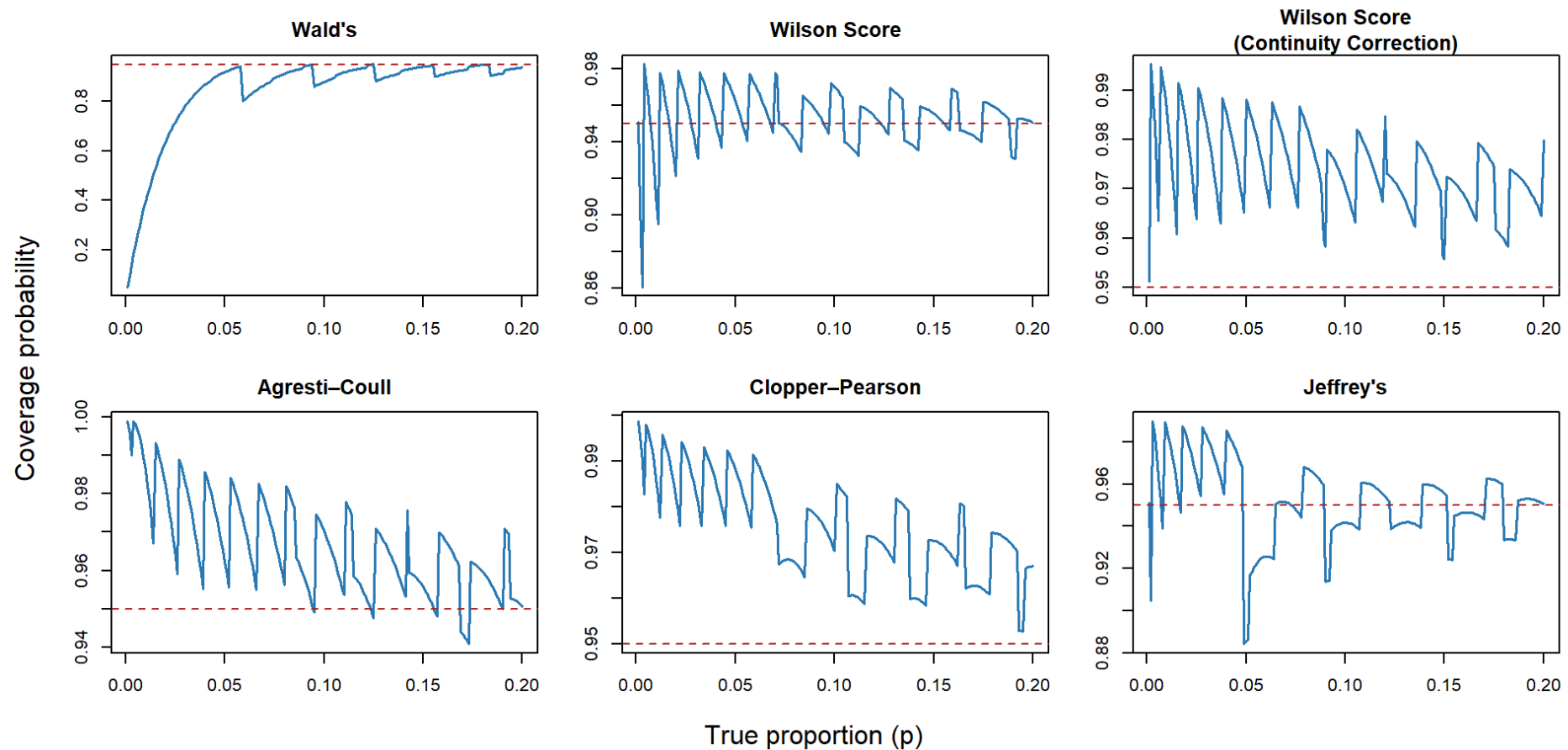
AMS Subject Classification:

- 62F25, 62F40.

Pires & Amado (2008): *Interval estimators for a binomial proportion: Comparison of twenty methods*

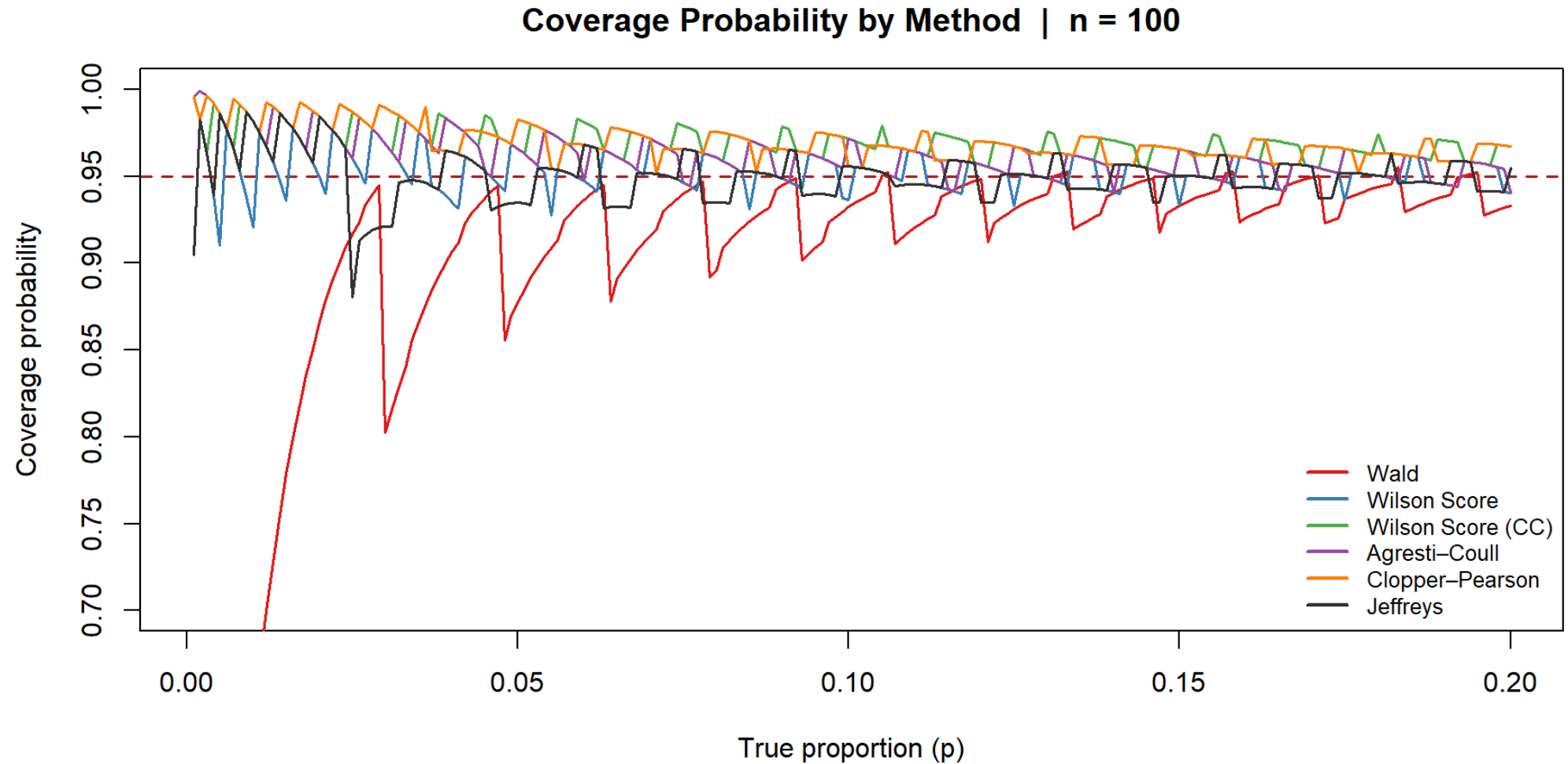
- Behaviour evaluated under:
 - Coverage probability
 - Average expected 95% CI Width
 - Under different sample sizes

Coverage Probability at Low Event Proportions ($n = 50$)

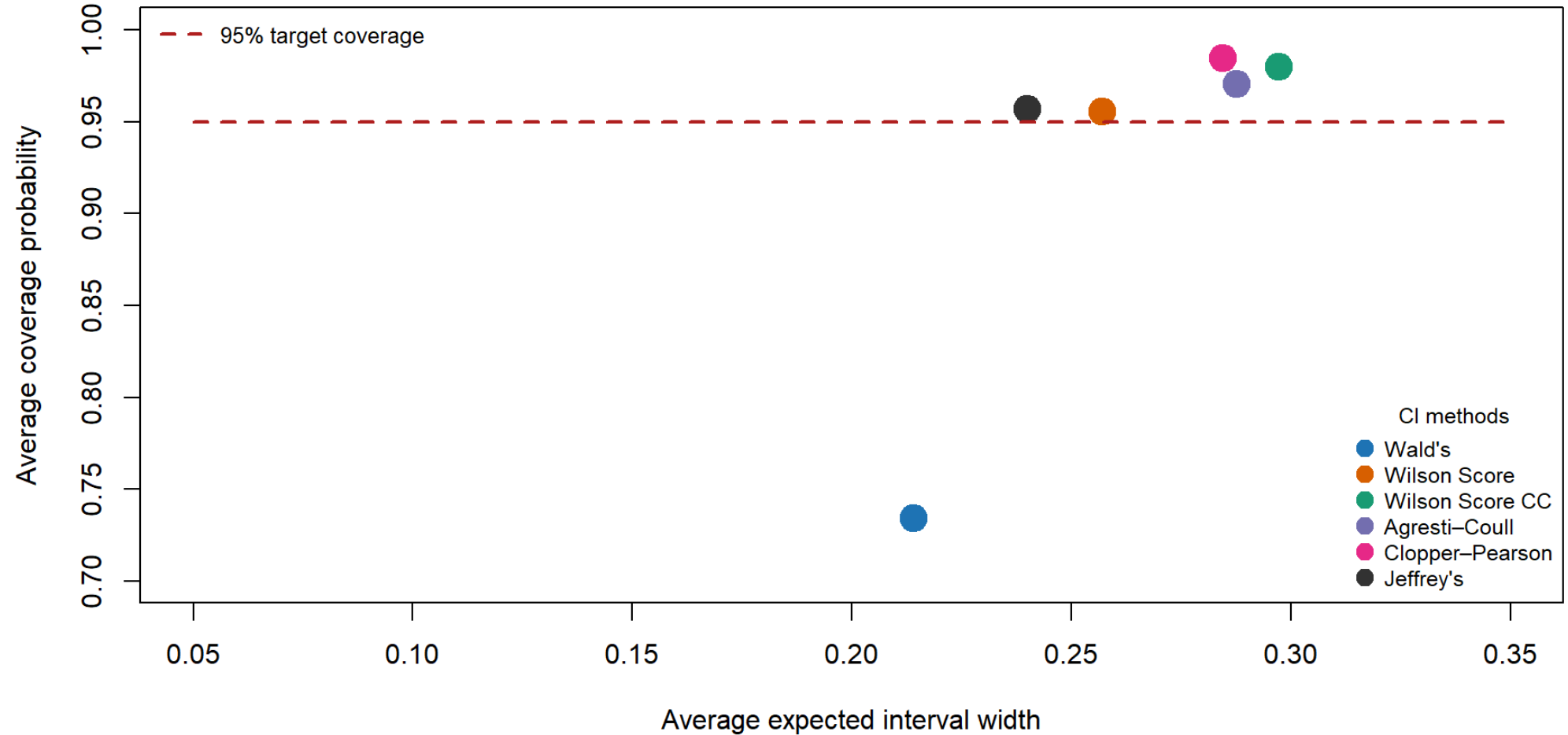


Dashed red = nominal 95%.

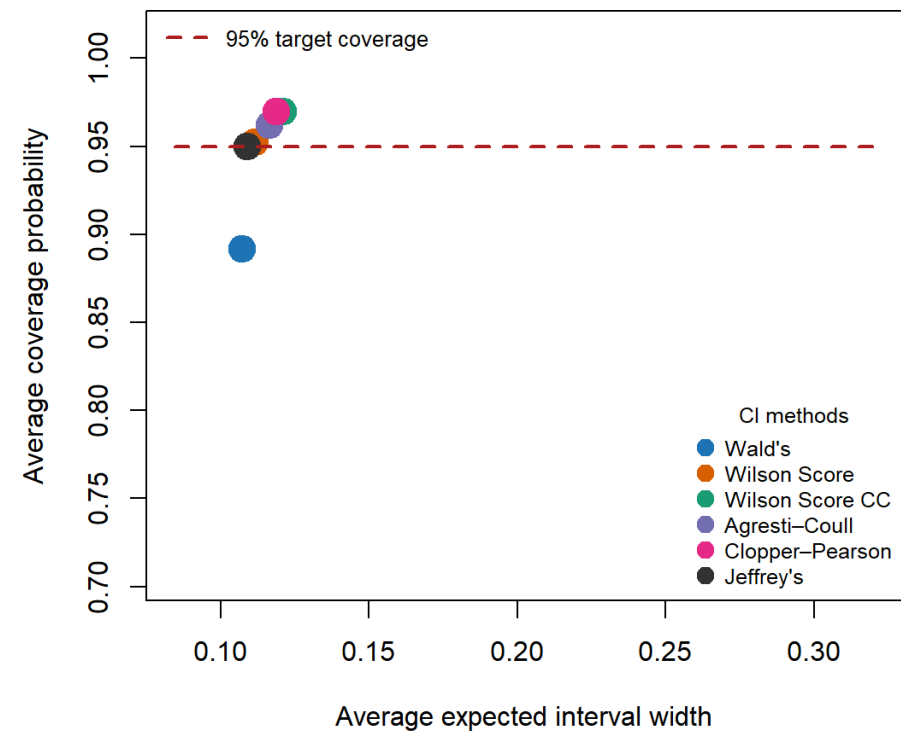
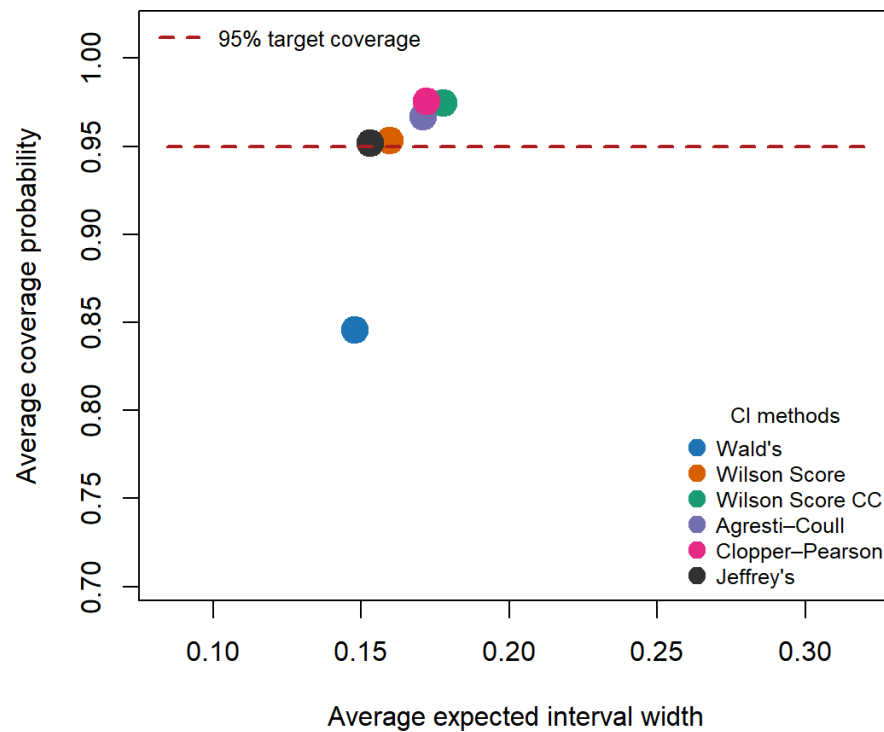
Coverage Probability $n = 100$



Expected 95% CI Width $n = 20$



Expected 95% CI Width: $n = 50$ vs $n = 100$



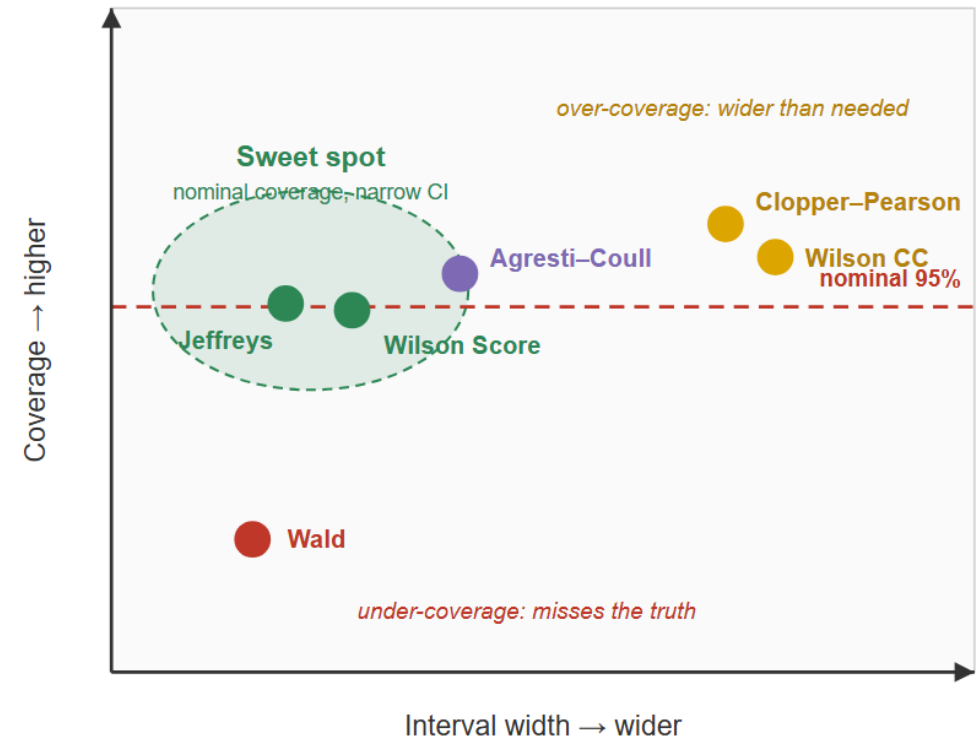
Simulation Study: Key Takeaways

Wald's interval fails at small proportions - undercoverage regardless of n

Clopper–Pearson & Wilson CC over-cover - unnecessarily wide intervals

Wilson Score & Jeffreys best balance coverage accuracy and interval width

Recommended: Wilson Score or Jeffreys for small proportion clinical endpoints



Sample Size: Safety Endpoint

- Proposed Safety Endpoint for Composite (Death, Amputation, TVR): $\leq 12\%$
- Approved Safety Endpoint for Composite (Death, Amputation, TVR): $\leq 11\%$
- Chosen Proportion Method: Exact Test

	Value
Significance (α)	0.025 (one-sided)
Power	90%
Approved Safety Endpoint Proportion (PG)	0.11
Assumed Safety Endpoint Proportion (p_1)	0.05

Software Comparison



NCSS PASS $n = 220$

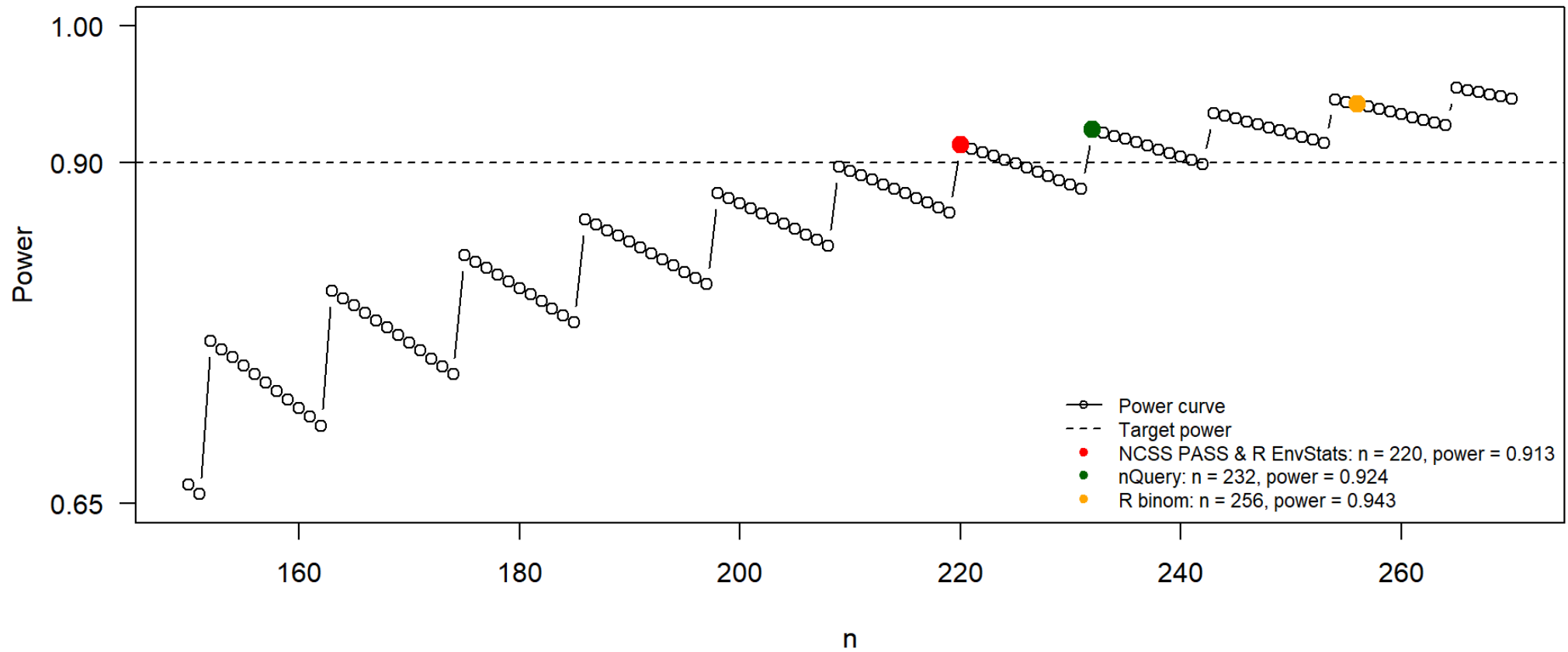


nQuery $n = 232$



R EnvStats $n = 220$ | **R binom** $n = 256$

One-sided sample size comparison for 30-day Composite
PG = 0.11 | Expected Composite = 0.05 | Target power = 0.9

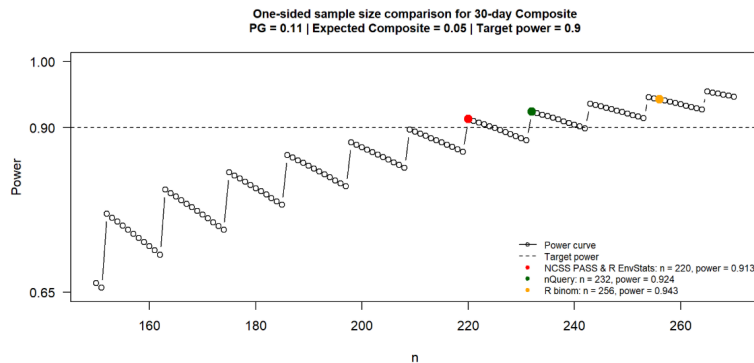


Why the saw-tooth shape?

It is a discrete fraction: $\frac{\text{critical value}}{n}$, e.g.

$$\frac{1}{50}, \frac{1}{51}, \frac{1}{52}, \frac{2}{53}$$

- Sample size increases but the **critical value** remains constant → power declines
- Then the **critical value** jumps → fraction rises → easier again → power peaks.



Different Software:

- n=220: First N hitting 90% (but immediately drops)
- n=232: Power reliably above 90%
- n=256: Power never drops below 90% for any N

Shiny App

Github:

[*https://github.com/FilipMKgit/MDX2526*](https://github.com/FilipMKgit/MDX2526)

Posit Cloud:

[*https://xj73kq-filip0mikolaj-klosowski.shinyapps.io/shiny_app/*](https://xj73kq-filip0mikolaj-klosowski.shinyapps.io/shiny_app/)



How this relates to Roche?

- Every methodological choice has consequences
 - CI method → Can affect sample size & results
- Results must be **reproducible and transparent**
 - Statistical methods → Must be defined, justified & consistent
- Statistics must be **communicable** for a cross-functional team
 - No man is an island
- Never rely on a single implementation
 - **Cross-validation** is essential



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Thank you